Experiment -1

Python Program To Solve 8-Puzzle Problem

Code:

```
from collections import deque

def solve(start, goal):
    queue = deque([(start, [])])
    visited = set()

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        if state in visited:
            continue

        visited.add(state)
        zero = state.index(0)

        moves = {
            0: [1, 3], 1: [0, 2, 4], 2: [1, 5],
            3: [0, 4, 6], 4: [1, 3, 5, 7],
            5: [2, 4, 8], 6: [3, 7],
            7: [4, 6, 8], 8: [5, 7]
        }
```

```
        for i in moves[zero]:
            new = list(state)
            new[zero], new[i] = new[i], new[zero]
            new = tuple(new)

            if new not in visited:
                queue.append((new, path + [state]))

    return None

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

solution = solve(start, goal)

if solution:
    for state in solution:
        for i in range(0, 9, 3):
            print(state[i:i+3])
        print()
else:
    print("No solution")
```

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        for i in moves[zero]:
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```python
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```

```
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)



...Program finished with exit code 0
Press ENTER to exit console.
```